

THE NISAN–SZEGEDY BOUND ON RELEVANT VARIABLES IS NEVER TIGHT, AND $R_3 = 10$

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ABSTRACT. Nisan and Szegedy proved in 1994 that a Boolean function whose real multilinear degree is at most d depends on at most $d2^{d-1}$ variables. Writing R_d for the maximum number of relevant variables of a degree- d Boolean function, this says $R_d \leq d2^{d-1}$, and the bound is attained for $d = 1$ and $d = 2$. We prove two results. First, the Nisan–Szegedy bound is never attained for $d \geq 3$: $R_d \leq d2^{d-1} - 1$ for every $d \geq 3$. The proof shows that in a hypothetical extremal function every derivative is a character times the indicator of an affine subspace, that every vertex link is the facet complex of a cross-polytope, and that the resulting term hypergraph splits into at least two cross-polytope boundaries on disjoint variable sets, which no Boolean function admits. Second, we determine R_3 exactly: $R_3 = 10$, attained by the Chiarelli–Hatami–Saks function Ξ_3 . Consequently $R_1 = 1$, $R_2 = 4$, $R_3 = 10$, and $22 \leq R_4 \leq 31$. The value $R_3 = 10$ is confirmed independently by an exhaustive canonical-form search over a finite reformulation of the problem, and the first structural steps of the argument have been formalised in Lean 4.

1. INTRODUCTION

1.1. The problem. Let $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ be a Boolean function, written in the ± 1 convention, and let

$$f = \sum_{S \subseteq [n]} \widehat{f}(S) \chi_S, \quad \chi_S(x) = \prod_{i \in S} x_i,$$

be its Fourier (real multilinear) expansion. The *degree* $\deg f$ is the largest $|S|$ with $\widehat{f}(S) \neq 0$. A variable x_i is *relevant* if f depends on it, that is, if $f(x) \neq f(x^{\oplus i})$ for some x , where $x^{\oplus i}$ flips the i -th coordinate; equivalently, if $\widehat{f}(S) \neq 0$ for some $S \ni i$.

Set

$$R_d := \sup\{n : \text{some } f: \{-1, 1\}^n \rightarrow \{-1, 1\} \text{ with } \deg f \leq d \text{ has all } n \text{ variables relevant}\}.$$

That R_d is finite is the theorem of Nisan and Szegedy [2]: a degree- d Boolean function is a $d2^{d-1}$ -junta, so

$$R_d \leq d2^{d-1}. \tag{1}$$

Chiarelli, Hatami and Saks [1] showed that the truth is exponential rather than $d2^{d-1}$: there is an absolute constant $C \leq 6.614$ with $R_d \leq C2^d$, and they gave, for each d , an explicit degree- d function Ξ_d with

$$3 \cdot 2^{d-1} - 2 \quad \text{relevant variables}, \tag{2}$$

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so that $R_d \geq 3 \cdot 2^{d-1} - 2$. Wellens [6] improved the constant to 4.394. Writing

$$C_{44} := \sup_{d \geq 1} \frac{R_d}{2^d},$$

the current state of the art is $3/2 \leq C_{44} \leq 4.394$; this is entry 44a on Tao's optimisation problems page [4].

The improved constants of [1, 6] are asymptotic. For small d they are worse than (1): the Chiarelli–Hatami–Saks optimisation attains its minimum at $d = 12$, and Wellens' bound beats $d 2^{d-1}$ only from roughly $d \geq 9$ on. For $d \leq 8$, therefore, (1) is still the best published upper bound, and the exact values of R_d have not been recorded anywhere for any $d \geq 3$: the problem page [4] lists $10 \leq R_3 \leq 12$ and states that exact values are not known.

1.2. Results. We settle the tightness of (1) for every d , and determine R_3 .

Theorem 1.1. *For every $d \geq 3$, no Boolean function of degree d has $d 2^{d-1}$ relevant variables. Hence*

$$R_d \leq d 2^{d-1} - 1 \quad (d \geq 3).$$

Theorem 1.2. $R_3 = 10$. *That is, a Boolean function of degree 3 has at most 10 relevant variables, and 10 is attained by the Chiarelli–Hatami–Saks function Ξ_3 .*

Theorem 1.1 is sharp in the sense that $d \geq 3$ cannot be weakened: for $d = 1$ the function $f(x) = x_1$ has $1 = 1 \cdot 2^0$ relevant variable, and for $d = 2$ the function

$$f(x_1, x_2, x_3, x_4) = \frac{1}{2} (x_1 x_2 + x_1 x_3 + x_2 x_4 - x_3 x_4) \quad (3)$$

is Boolean of degree 2 with $4 = 2 \cdot 2^1$ relevant variables. So the Nisan–Szegedy bound is attained exactly for $d \in \{1, 2\}$.

Combining the two theorems with (2) and (1), the exact state of knowledge for small d is now

$$R_1 = 1, \quad R_2 = 4, \quad R_3 = 10, \quad 22 \leq R_4 \leq 31,$$

the upper bound 31 for $d = 4$ being $4 \cdot 2^3 - 1$ from Theorem 1.1.

1.3. Method. Both theorems run on the same structural device, which we call the *link of a derivative*. If f has degree d and x_i is relevant, then the discrete derivative

$$D_i f(x) := \frac{f(x^{i \rightarrow 1}) - f(x^{i \rightarrow -1})}{2} = \sum_{S \ni i} \widehat{f}(S) \chi_{S \setminus i}$$

is a nonzero $\{-1, 0, 1\}$ -valued function of degree at most $d - 1$, and by granularity all of its Fourier coefficients are integer multiples of 2^{1-d} . Being $\{-1, 0, 1\}$ -valued is a very rigid constraint on a sparse, granular Fourier expansion. Lemma 3.1 makes this precise at the extreme end: if the Fourier mass of $D_i f$ is as small as it can possibly be, then the support of $D_i f$ is an affine subspace of $\mathbb{F}_2^n$ and all its coefficients are $\pm 2^{1-d}$, i.e. $D_i f$ is a character times the indicator of an affine subspace of codimension $d - 1$. A counting argument (Lemma 3.3) turns the affine subspace into the facet family of a cross-polytope when all the support sets have the same size, and a closure argument propagates the cross-polytope from one vertex to its neighbours until the whole term hypergraph is a disjoint union of 2^{d-2} cross-polytope boundaries. For $d \geq 3$ there are at least two of them, and a sum of two non-constant functions on disjoint variable sets is never ± 1 -valued (Lemma 3.5). That is Theorem 1.1.

For Theorem 1.2 the same device is pushed through a finer case analysis. Part I ($R_3 \leq 11$, Section 4) is the specialisation of the above to $d = 3$, where the cross-polytope is an octahedron and the term hypergraph is a closed triangulated surface with all vertex degrees 4. Part II

($R_3 \leq 10$, Section 5 and Appendix A) has an extra unit of slack in the influence budget, which allows the function to carry terms of degree below the top level and allows two vertices to have larger derivative mass. The slack is accounted for by a single identity (Section 5.3), and the resulting finitely many configurations are eliminated one by one.

1.4. Verification. The main theorems are proved by hand. Independently of the proofs, we reformulate “no degree-3 Boolean function has n relevant variables” as a purely finite arithmetic statement $F(n)$ about integer vectors (Section 6.1), and run an exhaustive orderly search over canonical forms. The search finds no solution for $n = 11$ and exactly one orbit for $n = 10$, namely Ξ_3 . A separate SAT-based structure search by an independent referee finds the same short list of candidate term structures and no Boolean sign vector on any of them. Finally, the first structural steps of Part I have been formalised in Lean 4 and kernel-checked on the standard axioms; we are explicit in Section 6.4 about what is and what is not machine-checked (the finite statements $F(11)$ and $F(12)$ are *not*).

1.5. Related work and novelty. Nisan and Szegedy [2] do not discuss whether (1) is tight for $d \geq 3$. Chiarelli–Hatami–Saks [1] and Wellens [5, 6] beat $d2^{d-1}$ only for large d , and neither states an exact value of R_3 or R_4 nor addresses tightness at small d ; Wellens’ Table 1 tabulates linear-programming bounds on block sensitivity, not on the number of relevant variables. A literature sweep carried out for this project found no paper, preprint, thesis or problem-tracker entry recording an exact value of R_d for any $d \geq 3$. The acknowledgements of [1] thank Avishay Tal for sharing exhaustive-search code for Boolean functions on few variables, but no results of such a search are reported there. To the best of our knowledge, Theorems 1.1 and 1.2 are new.

1.6. Organisation. Section 2 fixes notation and collects the standard facts we use. Section 3 proves Theorem 1.1. Section 4 proves $R_3 \leq 11$, Section 5 sets up the eleven-variable analysis and Appendix A carries it out in full, completing Theorem 1.2. Section 6 describes the computational verification. Section 7 collects open problems.

2. PRELIMINARIES

2.1. Fourier analysis on the cube. We use the ± 1 convention throughout: a Boolean function is a map $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$, and every $g: \{-1, 1\}^n \rightarrow \mathbb{R}$ has a unique multilinear expansion $g = \sum_{S \subseteq [n]} \hat{g}(S) \chi_S$ with $\chi_S(x) = \prod_{i \in S} x_i$. We identify a subset $S \subseteq [n]$ with its indicator vector in $\mathbb{F}_2^n$, so that $\chi_S \chi_T = \chi_{S \Delta T}$, where Δ is symmetric difference, i.e. addition in $\mathbb{F}_2^n$. Expectations $\mathbf{E}$ and probabilities $\mathbf{Pr}$ are with respect to the uniform measure on $\{-1, 1\}^n$, and Parseval reads $\mathbf{E}[g^2] = \sum_S \hat{g}(S)^2$; in particular $\sum_S \hat{f}(S)^2 = 1$ for Boolean f .

2.2. Granularity.

Fact 2.1 (Granularity; [3, Ex. 1.11], [2]). *If $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ has $\deg f \leq d$, then every Fourier coefficient $\hat{f}(S)$ is an integer multiple of 2^{1-d} .*

[todo: confirm the exercise number in O’Donnell’s book; the statement itself is standard and is already implicit in [2].]

Granularity is what makes the whole argument finite: for $\deg f \leq d$ the rescaled coefficients $n_S := 2^{d-1} \hat{f}(S)$ are integers with $\sum_S n_S^2 = 4^{d-1}$, and the problem becomes one about integer vectors.

2.3. Derivatives, influences, links. For $i \in [n]$ put

$$D_i f(x) := \frac{f(x^{i \rightarrow 1}) - f(x^{i \rightarrow -1})}{2} = \sum_{S \ni i} \hat{f}(S) \chi_{S \setminus i},$$

where $x^{i \rightarrow b}$ is x with the i -th coordinate set to b . If f is Boolean then $D_i f$ is $\{-1, 0, 1\}$ -valued, and $\deg D_i f \leq \deg f - 1$. The *influence* of i is

$$\mathbf{Inf}_i(f) = \Pr[f(x) \neq f(x^{\oplus i})] = \Pr[D_i f \neq 0] = \sum_{S \ni i} \hat{f}(S)^2,$$

and the variable x_i is relevant exactly when $\mathbf{Inf}_i(f) > 0$.

Fact 2.2 (Support bound; [2]). *A nonzero function $g: \{-1, 1\}^n \rightarrow \mathbb{R}$ of degree at most k is nonzero on at least a 2^{-k} fraction of the cube.*

Proof. Restrict to a subcube on which a top-degree monomial of g survives and induct on k ; this is the Schwartz–Zippel-type bound used in [2]. $\square$

Corollary 2.3. *If $\deg f \leq d$ and x_i is relevant, then $\mathbf{Inf}_i(f) \geq 2^{1-d}$.*

Proof. $D_i f$ is nonzero, $\{-1, 0, 1\}$ -valued and of degree at most $d - 1$, so by Fact 2.2 $\Pr[D_i f \neq 0] \geq 2^{1-d}$. $\square$

Fact 2.4 (Total influence). $\sum_{i=1}^n \mathbf{Inf}_i(f) = \sum_S |S| \hat{f}(S)^2 \leq \deg f$ for Boolean f .

Corollary 2.3 and Fact 2.4 give (1) at once: $d \geq \sum_i \mathbf{Inf}_i \geq n 2^{1-d}$, so $n \leq d 2^{d-1}$. Everything below is an analysis of the near-equality case.

Definition 2.5 (Mass, support hypergraph, link). Let $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ have $\deg f \leq d$ and put $n_S := 2^{d-1} \hat{f}(S) \in \mathbb{Z}$. The *support* of f is $\mathcal{F} := \{S : n_S \neq 0\}$, regarded as a hypergraph on the vertex set $[n]$; when all its members have size d we call it *d -uniform*. For a vertex i we write

$$m_i := \sum_{S \ni i} n_S^2 = 4^{d-1} \mathbf{Inf}_i(f)$$

for the *mass at i* , and we call the signed family

$$\text{lk}(i) := \{(S \setminus i, n_S) : S \in \mathcal{F}, S \ni i\}$$

the *link* of i . We often abuse notation and call the underlying set family $\{S \setminus i : S \in \mathcal{F}, S \ni i\}$ the link as well; when $\mathcal{F}$ is d -uniform the link is a family of $(d-1)$ -sets and, for $d = 3$, a graph.

By definition $2^{d-1} D_i f = \sum_{S \ni i} n_S \chi_{S \setminus i}$ is $\{0, \pm 2^{d-1}\}$ -valued with $\sum_{S \ni i} n_S^2 = m_i$. The entire paper consists of extracting structure from that one sentence.

2.4. The Chiarelli–Hatami–Saks lower bound. For completeness we record the construction from [1, §3] at $d = 2, 3$. Put

$$\Xi_2(a, b, u, w) := \frac{a+b}{2} u + \frac{a-b}{2} w = \frac{au + bu + aw - bw}{2},$$

which is the degree-2 Boolean function (3) on 4 variables, and

$$\Xi_3(s, t, x, y) := \frac{s+t}{2} \Xi_2(x) + \frac{s-t}{2} \Xi_2(y),$$

where $x = (x_a, x_b, x_u, x_w)$ and $y = (y_a, y_b, y_u, y_w)$ are disjoint 4-tuples of variables. Expanding,

$$\Xi_3 = \frac{1}{4} \left[(s+t)(x_a x_u + x_b x_u + x_a x_w - x_b x_w) + (s-t)(y_a y_u + y_b y_u + y_a y_w - y_b y_w) \right],$$

a homogeneous cubic with 16 terms of coefficient $\pm \frac{1}{4}$ on $2+4+4 = 10$ relevant variables. Thus $R_3 \geq 10$. In general Ξ_d has $3 \cdot 2^{d-1} - 2$ relevant variables, giving (2).

3. THE NISAN–SZEGEDY BOUND IS NEVER ATTAINED FOR $d \geq 3$

Throughout this section $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ has $\deg f \leq d$ with all n variables relevant, and we write

$$m := d - 1, \quad N := 2^m = 2^{d-1}.$$

3.1. Minimal mass forces an affine cube of unit terms.

Lemma 3.1 (Minimal support forces an affine cube of unit terms). *Let*

$$q = 2^{-m} \sum_{j=1}^J n_j \chi_{T_j}$$

with the $T_j \subseteq [n]$ distinct, the n_j nonzero integers, and $\sum_j n_j^2 = N$. Suppose q is $\{-1, 0, 1\}$ -valued and not identically 0. Then all $n_j = \pm 1$, $J = N$, the family $\{T_j\}$ is an affine subspace of dimension m of $\mathbb{F}_2^n$, and

$$n_j n_k n_l n_r = +1 \quad \text{whenever} \quad T_j \triangle T_k \triangle T_l \triangle T_r = \emptyset.$$

Proof. Put $s := 2^m q = \sum_j n_j \chi_{T_j}$, so s takes values in $\{0, \pm N\}$ and, by Parseval, $\mathbf{E}[s^2] = \sum_j n_j^2 = N$; hence $\mathbf{Pr}[s \neq 0] = 1/N$ and $\mathbf{E}[s^4] = N^2 \mathbf{E}[s^2] = N^3$. Expanding the fourth power and using $\mathbf{E}[\chi_{T_j} \chi_{T_k} \chi_{T_l} \chi_{T_r}] = \mathbf{1}\{T_j \triangle T_k \triangle T_l \triangle T_r = \emptyset\}$,

$$\mathbf{E}[s^4] = \sum_{\substack{(j,k,l,r) \\ T_j \triangle T_k \triangle T_l \triangle T_r = \emptyset}} n_j n_k n_l n_r.$$

For each ordered triple (j, k, l) there is *at most one* index r with $T_r = T_j \triangle T_k \triangle T_l$, because the T 's are distinct; call the set of triples possessing such an r the *dominant* set Dom , and write $r(j, k, l)$ for that index. Then

$$\begin{aligned} N^3 = \mathbf{E}[s^4] &\leq \sum_{(j,k,l) \in \text{Dom}} |n_j n_k n_l| |n_{r(j,k,l)}| \\ &\leq \sum_{(j,k,l) \in \text{Dom}} n_j^2 n_k^2 |n_l| |n_{r(j,k,l)}| \end{aligned} \tag{4}$$

$$\begin{aligned} &\leq \left(\sum_{\text{Dom}} n_j^2 n_k^2 n_l^2 \right)^{1/2} \left(\sum_{\text{Dom}} n_j^2 n_k^2 n_{r(j,k,l)}^2 \right)^{1/2} \\ &\leq (N^3)^{1/2} (N^3)^{1/2} = N^3. \end{aligned} \tag{5}$$

Line (4) uses $|n| \leq n^2$ for nonzero integers n , applied to the factors n_j and n_k ; line (5) is Cauchy–Schwarz. The final bound uses $\sum_{\text{Dom}} n_j^2 n_k^2 n_l^2 \leq (\sum_j n_j^2)^3 = N^3$ and, for the second factor, that for fixed (j, k) the map $l \mapsto r(j, k, l)$ is injective on Dom (since l is recovered from r by $T_l = T_j \triangle T_k \triangle T_r$), so that $\sum_l n_{r(j,k,l)}^2 \leq \sum_j n_j^2 = N$ and the whole sum is at most $(\sum_j n_j^2)^2 \cdot N = N^3$.

Equality throughout forces the following.

- (i) Dom is the set of *all* triples: the family $\{T_j\}$ is closed under $(x, y, z) \mapsto x \triangle y \triangle z$, hence is an affine subspace of $\mathbb{F}_2^n$, of some dimension k , and $J = 2^k$.

- (ii) Equality in (4) forces $|n_j n_k| = 1$ whenever the triple contributes, and equality in Cauchy–Schwarz gives $|n_{r(j,k,l)}| = \lambda |n_l|$ for all j, k, l and some $\lambda > 0$; taking $j = k$ gives $r = l$ and $\lambda = 1$, and for fixed k, l the map $j \mapsto r(j, k, l)$ is onto, so all $|n_j|$ equal a common value c . (In fact $c = 1$ already follows from (4); the argument is kept because it also delivers the count J .)
- (iii) All products $n_j n_k n_l n_{r(j,k,l)}$ are positive.

From (ii), $2^k c^2 = N = 2^m$, so $c = 2^t$ with $k = m - 2t$. If $t \geq 1$ then $|s| \leq c 2^k = 2^{m-t} < N$ pointwise, so $s \in \{0, \pm N\}$ forces $s \equiv 0$, contradicting $q \neq 0$. Hence $c = 1$, $k = m$ and $J = N$. $\square$

Remark 3.2. Item (iii) says that $\varepsilon_j := n_j$ is an affine character on the affine subspace $\{T_j\}$: writing $\{T_j\} = T_0 \triangle L$ for a linear subspace L of dimension m , we have $\varepsilon_{T_0 \triangle A} = \varepsilon_0 \psi(A)$ for a homomorphism $\psi: L \rightarrow \{\pm 1\}$. Consequently, for any basis $D_1, \dots, D_m$ of L ,

$$q = \pm \chi_{T_0} \prod_{i=1}^m \frac{1 + \psi(D_i) \chi_{D_i}}{2},$$

that is: a minimal-mass derivative is a character times the indicator of an affine subspace of codimension $m = d - 1$. The converse also holds, every such product being $\{-1, 0, 1\}$ -valued. Note that Lemma 3.1 uses no degree hypothesis at all; it is a statement about sparse, granular, $\{-1, 0, 1\}$ -valued functions, and it applies to any variable of minimal influence 2^{1-d} in any degree- d Boolean function, homogeneous or not.

3.2. Equal sizes force a cross-polytope.

Lemma 3.3 (Equal sizes force a cross-polytope). *Let $\{T_p\} = T_0 \triangle L$ be an affine subspace of dimension m of subsets of $[n]$, all of whose members have size exactly m . Then there are m pairwise disjoint pairs $\{t_i, t'_i\}$, $i = 1, \dots, m$, with $T_0 = \{t_1, \dots, t_m\}$ and*

$$\{T_p\} = \{ \{u_1, \dots, u_m\} : u_i \in \{t_i, t'_i\} \},$$

that is, $\{T_p\}$ is the facet family of an m -dimensional cross-polytope on the $2m$ vertices $t_1, t'_1, \dots, t_m, t'_m$.

Proof. For $A \in L \setminus \{\emptyset\}$ and any p , the equality $|T_p \triangle A| = m = |T_p|$ gives $|A| = 2|A \cap T_p|$. Applying this with $T_p = T_0 \triangle B$ for $B \in L$,

$$\frac{|A|}{2} = |(T_0 \triangle B) \cap A| = |T_0 \cap A| + |B \cap A| - 2|T_0 \cap A \cap B|,$$

and since $|A|/2 = |T_0 \cap A|$ we obtain

$$|A \cap B| = 2|T_0 \cap A \cap B| \quad \text{for all } A, B \in L.$$

The $\mathbb{F}_2$ -linear map $L \rightarrow 2^{T_0}$, $A \mapsto A \cap T_0$, is injective (if $A \cap T_0 = \emptyset$ then $|A| = 2|A \cap T_0| = 0$), and both spaces have dimension m , so it is a bijection. For $t \in T_0$ let $A_t \in L$ be the element with $A_t \cap T_0 = \{t\}$; then $|A_t| = 2$, say $A_t = \{t, t'\}$ with $t' \notin T_0$. For $t \neq s$ in T_0 , $|A_t \cap A_s| = 2|T_0 \cap A_t \cap A_s| = 0$, so the elements t' are distinct and the pairs $\{t, t'\}$ are disjoint. The A_t form a basis of L , and for $P \subseteq T_0$,

$$T_0 \triangle \left(\bigtriangleup_{t \in P} A_t \right) = (T_0 \setminus P) \cup \{t' : t \in P\},$$

which is exactly the asserted description. $\square$

Fact 3.4 (Antipodal pairs are readable from the facets). *The facet family of an m -dimensional cross-polytope determines its antipodal pairs: two vertices are antipodal if and only if they never occur in a common facet, every non-antipodal pair lying in a common facet.*

We use Fact 3.4 to read off the pairs of a link from its facets.

3.3. Disjoint sums.

Lemma 3.5 (Disjoint sums). *Let g and h be real functions of disjoint sets of variables with $g(x) + h(y) \in \{\pm 1\}$ for all x, y . If g is not constant then h is constant.*

Proof. Let $u \neq u'$ be two values of g . Then $h(y) \in \{1 - u, -1 - u\} \cap \{1 - u', -1 - u'\}$ for every y , and two distinct two-element sets meet in at most one point, so h is constant. $\square$

In particular a sum $f_1(x) + f_2(y)$ of two non-constant functions of disjoint variable sets is never ± 1 -valued, since by Lemma 3.5 one of them must be constant. We use this repeatedly, always in the following form: if the support of a Boolean f splits into two nonempty parts on disjoint variable sets, each carrying positive Fourier weight on non-constant characters, we have a contradiction.

3.4. Proof of Theorem 1.1.

Proof of Theorem 1.1. Let $d \geq 3$ and suppose $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ has $\deg f \leq d$ and $n = d2^{d-1}$ relevant variables.

Step 1: the extremal shape. By Corollary 2.3 and Fact 2.4,

$$d \geq \deg f \geq \sum_i \mathbf{Inf}_i(f) \geq n2^{1-d} = d,$$

so equality holds everywhere: $\deg f = d$ exactly, every $\mathbf{Inf}_i(f) = 2^{1-d}$, and $\sum_S |S| \widehat{f}(S)^2 = d = d \sum_S \widehat{f}(S)^2$, which forces all Fourier weight onto level d . Thus f is a homogeneous degree- d multilinear polynomial, $f = \sum_{|S|=d} c_S \chi_S$.

Step 2: every link is a cross-polytope. Fix a relevant variable i and write $D_i f = 2^{-m} \sum_{S \ni i} n_S \chi_{S \setminus i}$ with $n_S := 2^m c_S \in \mathbb{Z}$ by Fact 2.1. Then $\sum_{S \ni i} n_S^2 = 2^{2m} \mathbf{Inf}_i(f) = 2^m = N$, the function $D_i f$ is $\{-1, 0, 1\}$ -valued and nonzero, and the sets $S \setminus i$ (with i fixed) are distinct. Lemma 3.1 applies and gives: every n_S with $S \ni i$ equals ± 1 , so every coefficient of f is $\pm 2^{-m}$ and, by Parseval, f has exactly 2^{2m} terms, forming a d -uniform hypergraph $\mathcal{F}$ in which every vertex lies in exactly $N = 2^{d-1}$ terms; and the link $\text{lk}(i) = \{S \setminus i : S \in \mathcal{F}, S \ni i\}$ is an affine subspace of dimension m consisting of m -sets. By Lemma 3.3, $\text{lk}(i)$ is the facet family of an m -dimensional cross-polytope: there are m disjoint pairs $P_1(i), \dots, P_m(i)$ of vertices (the *antipodal pairs at i*), and the terms at i are exactly the sets $\{i\} \cup \{\text{one vertex from each } P_k(i)\}$. In particular two vertices antipodal at i never lie in a common term with i , and i has exactly $2m$ neighbours.

Step 3: closure. Fix a vertex v with pairs $P_k = \{t_k, t'_k\}$, $k = 1, \dots, m$; note $m \geq 2$ because $d \geq 3$. Put $a := t_1$. The terms at v containing a are the sets $\{v, a\} \cup (\text{one from each } P_2, \dots, P_m)$; removing a , these are 2^{m-1} facets of the cross-polytope $\text{lk}(a)$, all containing v . By Fact 3.4 the pairs of $\text{lk}(a)$ are therefore $P_2, \dots, P_m$ together with one further pair $\{v, w\}$ for a unique vertex w , the *antipode of v at a* . Clearly $w \notin \{v\} \cup P_2 \cup \dots \cup P_m$. Also $w \neq t'_1$: otherwise the term $\{a, t'_1, t_2 \text{ or } t'_2, \dots\}$ exists, and removing t_2 from it exhibits a facet of $\text{lk}(t_2)$ containing both $a = t_1$ and t'_1 — but P_1 is a pair of $\text{lk}(t_2)$ (as the terms at v containing t_2 show), so t_1 and t'_1 are antipodal at t_2 , and Fact 3.4 forbids their lying in a common facet. So w is a new vertex.

The terms at a are therefore $\{a\} \cup \{v \text{ or } w\} \cup (\text{one from each } P_2, \dots, P_m)$. In particular $\{a, w, t_2, u_3, \dots, u_m\}$ is a term for suitable $u_k \in P_k$, so removing t_2 gives a facet of $\text{lk}(t_2)$ containing a and w . The pairs of $\text{lk}(t_2)$ are $P_1, P_3, \dots, P_m$ together with $\{v, w_2\}$, where w_2 is the antipode of v at t_2 ; since w appears in a facet of $\text{lk}(t_2)$ and is none of the vertices of

$P_1, P_3, \dots, P_m$, we get $w = w_2$. Symmetrically the antipode of v at every t_k and every t'_k equals w . Hence w is adjacent to all $2m$ vertices of $P_1 \cup \dots \cup P_m$; these exhaust the $2m$ neighbours of w , and the terms at w are exactly $\{w\} \cup (\text{one from each } P_k)$.

Therefore the connected component of v in $\mathcal{F}$ is exactly the boundary complex of the d -dimensional cross-polytope on the $2d$ vertices $\{v, w\} \cup P_1 \cup \dots \cup P_m$: it has $2d$ vertices and 2^d facets, and no other term of f touches these vertices, since every one of them already lies in its full quota of 2^{d-1} terms.

Step 4: at least two components, and the contradiction. By Step 2, $\mathcal{F}$ has $2^{2m} = 4^{d-1}$ terms, and by Step 3 it is a disjoint union of cross-polytope boundaries, each with 2^d terms on its own $2d$ vertices. Hence there are exactly

$$\frac{2^{2m}}{2^d} = 2^{d-2} \geq 2 \quad (d \geq 3)$$

components, on pairwise disjoint variable sets. Writing $f = f_1 + g$, where f_1 collects the 2^d terms of the first component (on its $2d$ variables) and g the rest, both f_1 and g are non-constant, since each carries positive Fourier weight on non-constant characters. This contradicts Lemma 3.5. Therefore no degree- d Boolean function has $d2^{d-1}$ relevant variables, and $R_d \leq d2^{d-1} - 1$ for $d \geq 3$. $\square$

Remark 3.6. The exponent $2^{d-2} \geq 2$ in Step 4 is exactly where $d \geq 3$ is used, and it is the only place. For $d = 2$ one has $2^{d-2} = 1$: the single component is the boundary of the 2-dimensional cross-polytope, i.e. a square, and the four edges with signs $(+, +, +, -)$ give the Boolean function (3) with $4 = 2 \cdot 2^1$ relevant variables. Step 3 also needs $m \geq 2$, i.e. $d \geq 3$, for the vertex t_2 to exist. So the proof correctly fails at $d = 2$, as it must.

4. PART I: NO DEGREE-3 BOOLEAN FUNCTION HAS 12 RELEVANT VARIABLES

This section proves the case $d = 3$ of Theorem 1.1 in a self-contained form, which is also the first half of Theorem 1.2. The reader who has read Section 3 may skip to Section 5; we keep the $d = 3$ argument because it runs through a closed-surface classification that is used again, in a more delicate form, in Part II.

Theorem 4.1. *No Boolean function of degree 3 has 12 relevant variables. Hence $R_3 \leq 11$.*

Proof. **Step 1: twelve relevant variables force a homogeneous cubic with 16 terms $\pm \frac{1}{4}$, 4-regular.** With 12 relevant variables, Corollary 2.3 and Fact 2.4 give $3 \geq \sum_i \mathbf{Inf}_i \geq 12 \cdot \frac{1}{4} = 3$. Hence every $\mathbf{Inf}_i = \frac{1}{4}$ exactly, and $\sum_S |S| \widehat{f}(S)^2 = 3 = 3 \sum_S \widehat{f}(S)^2$, so all Fourier weight sits on level 3: $f = \sum_{|S|=3} c_S \chi_S$. By Fact 2.1, $n_S := 4c_S$ is an integer, $\sum_S n_S^2 = 16$ by Parseval, and $\mathbf{Inf}_i = \sum_{S \ni i} c_S^2 = \frac{1}{4}$ gives $m_i = \sum_{S \ni i} n_S^2 = 4$ for every vertex i . So at each vertex there is either one term with $|n_S| = 2$, or four terms with $|n_S| = 1$.

Suppose some S had $|n_S| = 2$. Then the three vertices of S lie in no other term, so $f = \pm \frac{1}{2} \chi_S + g$ with g a function of the other 9 variables. Then $f^2 = \frac{1}{4} + g^2 \pm \chi_S g = 1$; the monomials of $\chi_S g$ all contain S while those of $\frac{3}{4} - g^2$ avoid S , so $\chi_S g = 0$, i.e. $g = 0$, contradicting $\sum \widehat{g}(\cdot)^2 = \frac{3}{4}$. Hence all $n_S = \pm 1$:

$$f = \frac{1}{4} \sum_{S \in \mathcal{F}} \varepsilon_S \chi_S,$$

with $\mathcal{F}$ a 3-uniform hypergraph on the 12 vertices, $|\mathcal{F}| = 16$, and every vertex in exactly 4 triples.

Step 2: every vertex link is a 4-cycle. Fix a vertex i . Then $D_i f = \sum_{S \ni i} c_S \chi_{S \setminus i} = \frac{1}{4} \sum_{j=1}^4 \varepsilon_j \chi_{T_j}$, where $T_j = S_j \setminus \{i\}$ are four distinct pairs. Since $D_i f$ is $\{-1, 0, 1\}$ -valued, writing $u_j = \varepsilon_j \chi_{T_j}(x) \in \{\pm 1\}$ we need $\sum_j u_j \in \{0, \pm 4\}$ for every x , i.e. never ± 2 , i.e. the number of j with $u_j = -1$ is always even, i.e. $\prod_j u_j \equiv 1$. That is, $(\prod_j \varepsilon_j) \chi_{T_1 \triangle T_2 \triangle T_3 \triangle T_4} \equiv 1$, so

$$T_1 \triangle T_2 \triangle T_3 \triangle T_4 = \emptyset \quad \text{and} \quad \prod_j \varepsilon_j = +1.$$

Four distinct pairs with empty symmetric difference: every vertex lies in an even number of them, there are 8 incidences, and a vertex lying in all four would leave the other four endpoints in one pair each (impossible, as those pairs would then have odd degrees). So exactly four vertices lie in exactly two pairs each, i.e. the pairs form the unique 2-regular simple graph on 4 vertices, a 4-cycle $a-b-c-d-a$. Consequently the four triples at i are $\{i, a, b\}, \{i, b, c\}, \{i, c, d\}, \{i, d, a\}$: the link of i in $\mathcal{F}$ is C_4 , and every pair $\{i, x\}$ lies in exactly 2 triples (for $x \in \{a, b, c, d\}$) or in none.

Step 3: $\mathcal{F}$ is a closed triangulated surface with all vertex degrees 4, hence two disjoint octahedra. By Step 2 the pure 2-dimensional complex $|\mathcal{F}|$ has every edge in exactly two triangles and every vertex link a single cycle C_4 , so it is a closed 2-manifold (possibly disconnected), triangulated with $V = 12$ vertices, $F = 16$ faces and $E = (3 \cdot 16)/2 = 24$ edges. Its Euler characteristic is $V - E + F = 12 - 24 + 16 = 4$. A connected closed surface has $\chi \leq 2$, so there are at least two components. Each component is a closed surface all of whose vertices have degree 4, so with V' vertices it has $E' = 2V'$ and $F' = 4V'/3$, whence $\chi' = V' - 2V' + 4V'/3 = V'/3 \in \{1, 2\}$. The value $\chi' = 1$ would need $V' = 3$, impossible for a complex with a degree-4 vertex (and a triangulated projective plane needs at least 6 vertices in any case). So $\chi' = 2$ and $V' = 6$: the component is the octahedron, which is the unique 4-regular triangulated sphere (the complement of a 4-regular graph on 6 vertices is a perfect matching, so the graph is $K_{2,2,2}$ and all 8 faces are then forced).

There is also a topology-free route to the same conclusion: if $\text{lk}(i) = a-b-c-d$ then $\text{lk}(a) = b-i-d-x$ for some $x \neq c$ (were $x = c$, the link of b would contain a 3-cycle, impossible inside a C_4), and the closure forces the octahedron on $\{i, a, b, c, d, x\}$; this was checked exhaustively by an independent referee.

Two octahedra use all 12 vertices. Hence $\mathcal{F}$ is the set of faces of two vertex-disjoint octahedra and

$$f = f_1(x_1, \dots, x_6) + f_2(x_7, \dots, x_{12}),$$

with f_1, f_2 each carrying 8 terms $\pm \frac{1}{4}$ on disjoint variable sets, neither constant.

Step 4: a sum of two non-constant functions on disjoint variable sets is never Boolean. This is Lemma 3.5: if $f_1(x) + f_2(y) \in \{\pm 1\}$ for all x, y and f_1 takes two values $u \neq u'$, then $f_2(y) \in \{1 - u, -1 - u\} \cap \{1 - u', -1 - u'\}$ for all y , an intersection with at most one element, so f_2 is constant — contradicting that f_2 has Fourier weight $\frac{1}{2}$ on non-constant characters.

Equivalently and more directly: for disjoint triples S in the first octahedron and T in the second, the coefficient of $\chi_{S \cup T}$ in f^2 is $2\varepsilon_S \varepsilon_T / 16 \neq 0$, since no other pair of triples of $\mathcal{F}$ has union $S \cup T$; but $f^2 \equiv 1$, so every non-constant coefficient of f^2 vanishes. Contradiction. $\square$

5. PART II: THE ELEVEN-VARIABLE CASE

Throughout this section $f: \{-1, 1\}^{11} \rightarrow \{-1, 1\}$ has $\deg f \leq 3$ and all 11 variables relevant; we derive a contradiction. Write $f = \sum_S c_S \chi_S$ and $n_S := 4c_S \in \mathbb{Z}$ by Fact 2.1, so that

$\sum_S n_S^2 = 16$ by Parseval, and for a vertex i put

$$m_i := \sum_{S \ni i} n_S^2 = 16 \mathbf{Inf}_i(f) = 16 \mathbf{Pr}[D_i f \neq 0].$$

Compared with Part I there is one unit of slack in the influence budget: eleven variables need total influence at least $11/4$, while 3 is available. The two lemmas below say exactly how that slack can be spent, and the bookkeeping identity of Section 5.3 turns it into a finite list of configurations, which Appendix A eliminates.

5.1. The masses.

Lemma 5.1 (Values of m_i). $m_i \in \{4, 6, 8\}$ for every i , and $\sum_i m_i = \sum_S |S| n_S^2 \leq 48$.

Proof. The derivative $D_i f = \sum_{S \ni i} (n_S/4) \chi_{S \setminus i}$ is $\{-1, 0, 1\}$ -valued (because f is Boolean) and of degree at most 2, so $4D_i f = \sum_j n_j \chi_{T_j}$ takes values in $\{0, \pm 4\}$, where the T_j are distinct sets of size at most 2 and $\sum_j n_j^2 = m_i$. A relevant variable has $\mathbf{Inf}_i \geq \frac{1}{4}$ by Corollary 2.3, so $m_i \geq 4$. Also $\sum_i m_i = \sum_S |S| n_S^2 \leq 3 \sum_S n_S^2 = 48$, so with 11 vertices each of mass at least 4 we get $m_i \leq 48 - 40 = 8$ for every i .

It remains to exclude $m_i = 5$ and $m_i = 7$.

- Any pattern with an odd number of odd n_j has odd value $\sum_j n_j \chi_{T_j}(x)$ at every point, hence never lies in $\{0, \pm 4\}$. This kills five ± 1 's, seven ± 1 's, $\{\pm 2, \pm 1\}$ and $\{\pm 2, \pm 1, \pm 1, \pm 1\}$.
- The pattern $\{\pm 2, \pm 1, \pm 1\}$ (mass 6): $2u_0 + u_1 + u_2 \in \{0, \pm 4\}$ with $u_k \in \{\pm 1\}$ forces $u_1 = u_2$ for every x , i.e. $\chi_{T_1} \equiv \pm \chi_{T_2}$, impossible for distinct $T_1 \neq T_2$. So mass 6 occurs only as six terms ± 1 .

Since $m_i \leq 8$ forces $|n_j| \leq 2$ throughout, the listed patterns are all the ones that occur, and $m_i \in \{4, 6, 8\}$. $\square$

Remark 5.2. A shorter route to the parity part: $4D_i f$ is $\{0, \pm 4\}$ -valued and $\sum_j n_j \equiv \sum_j n_j^2 = m_i \pmod{2}$, so m_i is even.

5.2. The link shapes.

Lemma 5.3 (Link structure). *With the link of i as in Definition 2.5, written $\{T_j\} = \{S \setminus i : n_S \neq 0, S \ni i\}$ with signs ε_j :*

- (a) $m_i = 4$: four terms ± 1 ; the condition is necessary and sufficient that $T_1 \triangle T_2 \triangle T_3 \triangle T_4 = \emptyset$ and $\prod_j \varepsilon_j = +1$ (a sum of four ± 1 's avoids ± 2 iff the number of -1 's is even iff $\prod_j u_j \equiv 1$). Four distinct sets of size at most 2 in which every element lies in an even number of members are exactly:

- (L1) a 4-cycle of pairs on 4 vertices;
- (L2) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$;
- (L3) $\{\{a\}, \{b\}, \{a, c\}, \{b, c\}\}$;
- (L4) $\{\emptyset, \{a, b\}, \{b, c\}, \{a, c\}\}$.

(Enumeration, verified by computer: without $\emptyset$, four pairs with all degrees even form a C_4 ; singletons come in pairs $\{a\}, \{b\}$ needing $\{a, b\}$ plus a fourth set — $\emptyset$ gives (L2) — or $\{a, c\}, \{b, c\}$ giving (L3); with $\emptyset$ present the other three sets have symmetric difference $\emptyset$, so they are three pairs forming a triangle, giving (L4), or $\{a\}, \{b\}, \{a, b\}$, giving (L2) again.) In terms of f : (L1) means i lies in four cubic terms only; (L2) means a linear term $\{i\}$, quadratics $\{i, a\}, \{i, b\}$ and a cubic $\{i, a, b\}$; (L3) means quadratics $\{i, a\}, \{i, b\}$ and cubics $\{i, a, c\}, \{i, b, c\}$; (L4) means a linear term $\{i\}$ and cubics $\{i, a, b\}, \{i, b, c\}, \{i, a, c\}$. In particular: a link containing a singleton is (L2) or (L3);

- a link containing $\emptyset$ is (L2) or (L4); a link containing a triangle of pairs is (L4), a C_4 having no triangle.
- (b) $m_i = 6$: six terms ± 1 with $T_1 \triangle \cdots \triangle T_6 = \emptyset$ and $\prod_j \varepsilon_j = -1$ (six ± 1 's avoid ± 2 and ± 6 iff the number of -1 's is odd).
 - (c) $m_i = 8$: either two terms ± 2 (on any two distinct sets), or one ± 2 and four ± 1 's whose sets have symmetric difference $\emptyset$ and sign product -1 , or eight terms ± 1 with symmetric difference $\emptyset$, sign product $+1$, and never all eight values agreeing.
 - (d) A vertex whose only term has $|n_S| = 2$ is impossible, since then $D_i f = \pm \frac{1}{2} \chi_{S \setminus i}$ is not $\{-1, 0, 1\}$ -valued. Consequently a term with $|n_S| = 2$ forces every vertex of S to have mass 8 (mass 6 cannot contain a ± 2 , by the proof of Lemma 5.1).
 - (e) In a link consisting of pairs only (no lower-order term at i), the multiplicity $\lambda_{ix} := \#\{S \in \mathcal{F} : S \ni i, x\}$ equals the degree of x in the link graph; for (L1) it is 2 for the four link vertices and 0 otherwise.

Proof. Each part is the value-set analysis of $4D_i f = \sum_j n_j \chi_{T_j}$ carried out as in Lemma 5.1 and Step 2 of Theorem 4.1: the sum of k terms ± 1 lies in $\{0, \pm 4\}$ at every point iff the symmetric difference of the T_j is empty with the stated sign product, and the enumeration of admissible set families of size 4 is the elementary parity argument displayed in (a). Part (d) is immediate, and its consequence follows from Lemma 5.1. Part (e) is a restatement of the definition. $\square$

5.3. Bookkeeping. Put

$$\delta := \sum_S (3 - |S|) n_S^2 \quad (\text{the Fourier weight below the top level, weighted by codegree}), \quad e := \sum_i (m_i - 4).$$

Then $\sum_i m_i = \sum_S |S| n_S^2 = 48 - \delta$ and $\sum_i m_i = 44 + e$, so

$$e + \delta = 48 - 44 = 4, \quad e \geq 0, \quad \delta \geq 0. \quad (6)$$

The contributions to δ are: a quadratic term with $|n_S| = 1$ contributes 1; a quadratic with $|n_S| = 2$ contributes 4; a linear term ± 1 contributes 2; the constant term ± 1 contributes 3. A linear or constant term with $|n_S| = 2$ would contribute 8 or $12 > 4$ and is therefore impossible.

By Lemma 5.1, $e \in \{0, 2, 4\}$, with $e = 0$ meaning all $m_i = 4$, $e = 2$ meaning exactly one vertex of mass 6, and $e = 4$ meaning either two vertices of mass 6 or one of mass 8. Correspondingly $\delta \in \{4, 2, 0\}$.

5.4. A surface fact.

Fact 5.4 (Surfaces of non-positive Euler characteristic). *In a closed connected triangulated surface with V' vertices, E' edges, F' faces and Euler characteristic χ' , the relations $3F' = 2E'$ and $V' - E' + F' = \chi'$ give the degree sum $2E' = 6V' - 6\chi'$. Hence $\chi' \leq 0$ forces average degree at least 6. In every complex arising below, all degrees are 4 except at most two vertices of degree at most 8, so the average degree is at most $4 + 8/V' < 6$ as soon as $V' \geq 5$; and $V' \geq 5$ holds because some vertex has degree at least 4 (a closed triangulated surface has no vertex of degree less than 3, and $V' \geq \deg + 1$). Therefore every component occurring below is a sphere (degree sum $6V' - 12$) or a projective plane (degree sum $6V' - 6$, requiring at least 6 vertices, by Step 3 of Theorem 4.1).*

Fact 5.4 was checked by computer: for every degree pattern used in Appendix A, the relation $V' = 3\chi' + c/2$ (with c the excess degree) was tabulated, and every row with $\chi' \leq 0$ gives $V' \geq 2$.

5.5. Statement and proof of Part II.

Theorem 5.5. *No Boolean function of degree 3 has 11 relevant variables.*

Proof. By (6) we have $\delta \in \{4, 2, 0\}$. The three cases are eliminated in Appendix A: $\delta = 4$ in Section A.1, $\delta = 2$ in Section A.2, and $\delta = 0$ in Section A.3. Since the three cases are exhaustive, no such f exists. $\square$

Proof of Theorem 1.2. The function Ξ_3 of Section 2 is Boolean of degree 3 with 10 relevant variables, so $R_3 \geq 10$. Theorem 4.1 rules out 12 relevant variables and Theorem 5.5 rules out 11. If some degree-3 Boolean function had $n \geq 12$ relevant variables then $n \leq 12$ by (1), so $n = 12$, contradicting Theorem 4.1. Hence $R_3 = 10$. $\square$

6. COMPUTATIONAL VERIFICATION

The proofs above are complete as mathematics. Because the case analysis of Appendix A is long, we also verified the conclusion $R_3 = 10$ by three independent computational routes: an exhaustive search over a finite reformulation, an independent SAT-based structure search performed by a referee, and a partial formalisation in Lean 4. This section states precisely what each covers.

6.1. A finite reformulation. Let $n \geq 1$. A *coefficient vector* is an integer-valued map $N: \{S \subseteq [n] : |S| \leq 3\} \rightarrow \mathbb{Z}$; write $n_S := N(S)$ and

$$f_N(x) := \frac{1}{4} \sum_S n_S \chi_S(x), \quad x \in \{-1, 1\}^n.$$

Definition 6.1 (The statement $F(n)$). $F(n)$ asserts that there is *no* coefficient vector N with

- (i) $\sum_S n_S^2 = 16$;
- (ii) for every nonempty $U \subseteq [n]$, $\sum_{\substack{(S,T) \text{ ordered} \\ S \Delta T = U}} n_S n_T = 0$;
- (iii) every $i \in [n]$ lies in some S with $n_S \neq 0$.

Conditions (i) and (ii) together say exactly that $f_N^2 \equiv 1$, i.e. that f_N is $\{-1, 1\}$ -valued of degree at most 3: since $\chi_S \chi_T = \chi_{S \Delta T}$, the expansion $f_N^2 = \frac{1}{16} \sum_{S,T} n_S n_T \chi_{S \Delta T}$ is identically 1 iff its $U = \emptyset$ coefficient is 16 and all others vanish. Condition (iii) says all n variables are relevant (a variable is relevant exactly when it appears in the Fourier support). Conversely, if $f: \{-1, 1\}^n \rightarrow \{-1, 1\}$ has degree at most 3 then Fact 2.1 makes $N := 4\hat{f}$ a coefficient vector with $f = f_N$, and (i)–(iii) hold. Hence:

$$F(n) \implies \text{no degree-3 Boolean function has } n \text{ relevant variables.}$$

In particular $F(11)$ implies $R_3 \leq 10$, and with Ξ_3 (which witnesses that $F(10)$ is false) it gives $R_3 = 10$.

The search below is allowed to use only the following elementary consequences of $f_N^2 \equiv 1$, each provable in two lines and each independent of the hand proof:

- (E1) For each i , $D_i f_N = \frac{1}{4} \sum_{S \ni i} n_S \chi_{S \setminus i}$ is $\{-1, 0, 1\}$ -valued, so $4D_i f_N$ takes values in $\{0, \pm 4\}$ at every point.
- (E2) Parseval: $\sum_{S \ni i} n_S^2 = 16 \mathbf{Pr}[D_i f_N \neq 0]$ is an integer in $[4, 16]$ for relevant i , and $\sum_i \sum_{S \ni i} n_S^2 = \sum_S |S| n_S^2 \leq 48$.
- (E3) Symmetries preserving (i)–(iii): permutations of $[n]$; negation of a variable i (which flips n_S for $S \ni i$); and the global sign $N \mapsto -N$.

Nothing beyond (E1)–(E3) — in particular no link classification from Section 5 — is assumed by the search.

6.2. Exhaustive canonical search. The search program processes vertices one at a time; processing v fixes every coefficient n_S with $v \in S$, that is, it chooses the link $L_v = 4D_v f_N = \sum_{S \ni v} n_S \chi_{S \setminus v}$, a degree-at-most-2 polynomial in the other variables. Four pruning rules are used, each a necessary condition implied by (E1)–(E3) on any genuine solution, so that discarding a branch violating one of them can never lose a solution:

- (P1) *Budget*: the accumulated $\sum n_S^2$ never exceeds 16, and $m_v \leq 16$; this is (i).
- (P2) *Bounded conditional values*: for a partial assignment of a subset A of the link’s variables, the terms confined to A must keep the partial value of L_v inside $[-4, 4]$; this is (E1) evaluated at every completion.
- (P3) *Weighted degree*: for a link variable w , $\sum_{T \ni w} |c_T| \leq 4$; this is (E1) applied one derivative further, $D_w L_v \in \{0, \pm 2, \pm 4\}$, so $|a - b|, |a + b| \leq 4$ forces $|a| + |b| \leq 4$.
- (P4) *Needs*: $\sum_{u \text{ unprocessed}} \max(0, 4 - m_u) \leq 3(16 - \text{used})$, since each unprocessed vertex needs mass at least 4 by (E2) and any future coefficient contributes its square to at most 3 vertices.

Symmetry is broken by a canonical-form test using only (E3): the global sign is fixed once at the top; fresh variables are introduced in increasing label order; and among all images of the current partial solution under the group generated by permutations of the fresh used variables, negations of fresh variables, and (when the fixed part of the link vanishes) global negation of the link, only the lexicographically minimal image is kept. Every genuine solution has some image satisfying all of these simultaneously, so rejecting non-canonical images discards only redundant representatives of an orbit. Consequently the raw solution counts below are already *orbit* counts. Condition (ii) is never pruned: at the last vertex the program evaluates $4f_N(x)$ at all 2^n points and requires the value to lie in $\{+4, -4\}$; this full, unpruned check is the only place a branch is accepted.

The $n = 11$ run was split into six shards by the residue of the index of the canonical top-level link modulo 6. Every shard enumerates the same full list of 368 canonical top-level links and recurses only on its own residue class, so the six shards partition the search exhaustively and disjointly; the enumeration order is deterministic because the program is single-threaded and depends only on n .

TABLE 1. Exhaustive canonical search for $F(n)$. “Solutions” are orbit counts under the symmetry reduction described above.

n	solutions (orbits)	leaves	nodes	canonical rejects	time
3	14	26	1,887	176	0.00 s
4	102	225	24,373	2,755	0.01 s
5	809	2,329	280,282	31,155	0.15 s
6	1,773	6,538	2,986,948	187,681	2.44 s
7	1,060	11,786	23,632,808	530,408	33.20 s
8	213	4,015	77,313,447	1,288,595	171.77 s
10	1	120	44,447,494	1,063,032	824.60 s
11 (6 shards)	0	16	96,822,409	3,747,449	≈ 2.1 CPU-h

Table 1 records the results. The case $n = 9$ was not run: it is not needed for the $F(10)$ and $F(11)$ claims. The value $n = 10$ giving exactly one orbit matches the known truth, and the recovered solution is 16 terms of coefficient ± 1 , i.e. exactly Ξ_3 up to the symmetry group; this

is a strong sanity check on the canonical-form machinery, which is the most intricate part of the program. The value $n = 11$ giving 0 across all six disjoint shards is the search’s verdict that $R_3 \leq 10$.

Every emitted solution at $n = 6, 7, 8, 10$ was re-checked by a from-scratch independent verifier which parses the solution, evaluates f_N by brute force at all 2^n points, and tests (i), (ii) and genuine relevance (two points differing only in coordinate i with different values of f_N , rather than mere appearance in the support): 3047 solutions checked, 0 failures. So no false positives are reported. We note the residual caveat explicitly: the soundness argument above shows the *pruning rules* cannot lose a solution, but it is not a line-by-line proof that the C implementation is correct, and a false negative at $n = 11$ caused by an implementation bug cannot be excluded by rerunning the same binary. An earlier version of the program had a genuine bug of exactly this kind (scratch buffers indexed by recursion depth only, hence aliased across vertex steps); it reported 0 solutions and 2 leaves at $n = 10$, where the corrected program reports 1 solution and 120 leaves. All figures reported here are from the corrected program, and the $n = 10$ agreement with Ξ_3 is the check that detects that failure mode.

6.3. Independent structure search. Independently of both the hand proof and the search above, a referee ran a CDCL SAT search (CaDiCaL via `pysat`) with blocking clauses over the *term structures* of an eleven-variable solution: the supports satisfying the link-consistency conditions, enumerated up to isomorphism, with the Boolean sign vectors on each enumerated afterwards. The complete run found exactly three isomorphism classes of link-consistent 16-term supports on 11 vertices and *zero* Boolean sign vectors on them:

- degree sequence 4^{11} , four quadratic and twelve cubic terms — the $\delta = 4$ case with weight partition $\{2, 2\}$ and an (L4) link, plus an octahedron;
- degree sequence 4^{11} , two linear and fourteen cubic terms — the $\delta = 4$ case with weight partition $\{1, 1, 1, 1\}$ and $|N| = 4$, plus an octahedron;
- degree sequence $(8, 4^{10})$, sixteen cubic terms — the $\delta = 0$ case with link $C_4 \cup C_4$, i.e. two octahedra glued at a vertex.

These are precisely the three branches that the hand proof closes with the disjoint-sum lemma (Lemma 3.5); every other branch is closed at the link level in Appendix A and is reported infeasible by the search. A later rerun restricted to the branch where the distinguished vertex has mass 4 reports $420 + 585$ labelled structures, 2 isomorphism classes and 0 Boolean sign vectors, consistent with the above; that rerun’s log ends inside the mass-6 branch without a totals line, so we quote the complete run for the totals.

6.4. Partial formalisation in Lean 4. The finite statement $F(n)$ was transcribed into Lean 4 (toolchain `leanprover/lean4:v4.23.0`) as a frozen definition, and the first structural steps toward $F(12)$ were proved against it. The following is the exact status.

What is machine-checked. Twenty-three theorems are proved and kernel-checked, with no `sorry`, no `admit`, no `native_decide`, no user axioms and no `unsafe/implemented_by/extern` escape hatches; an automated gate runs `lake build`, greps for all of those, and runs `#print axioms` on every listed theorem. Every one of the twenty-three depends only on the three standard Lean axioms `propext`, `Classical.choice`, `Quot.sound`. They cover:

- the basic identities ($f_N^2 = 16$ in integer form, the pointwise membership (E1), divisibility and lower bounds on the mass, $\sum_i m_i \leq 48$);
- Lemma 5.1 for $n \geq 11$ (mass in $\{4, 6, 8\}$), and that at least nine of the eleven vertices have mass 4;
- the bookkeeping identity (6) in the form $e + \delta = 48 - 4n$;

- the first half of Lemma 5.3(a): four distinct link sets of size at most 2, coefficients ± 1 , symmetric difference empty, sign product $+1$, and every vertex in an even number of the four sets;
- Step 1 of Theorem 4.1 for $n = 12$ (all masses 4, all terms cubic, all coefficients ± 1 , all link sets of size 2);
- Step 2 of Theorem 4.1: four distinct pairs with empty symmetric difference form a 4-cycle, hence for $n = 12$ every vertex link is a 4-cycle.

The strongest certified statement is: in any 12-variable solution of the frozen F , at every vertex v there are four distinct vertices p, q, r, s such that every support set at v , with v removed, is one of the pairs pq, qr, rs, sp .

What is not machine-checked. Neither $F(12)$ nor $F(11)$ is formalised. The remaining steps for $F(12)$ are the octahedral closure (Step 3 of Theorem 4.1) and the disjoint-sum contradiction (Step 4); for $F(11)$, the whole of Appendix A. Mathlib has no classification of closed surfaces, so the closure arguments would have to be replaced by their topology-free finite versions, and the exhaustive search of the previous subsection is far beyond kernel `decide`. Theorem 1.1 is not formalised at all.

7. CONCLUDING REMARKS AND OPEN PROBLEMS

7.1. The state of the small-degree values. We now know

$$R_1 = 1, \quad R_2 = 4, \quad R_3 = 10, \quad 22 \leq R_4 \leq 31,$$

where $R_4 \geq 22 = 3 \cdot 2^3 - 2$ comes from Ξ_4 and $R_4 \leq 31 = 4 \cdot 2^3 - 1$ from Theorem 1.1. Determining R_4 is the obvious next question. The gap is wide, and the hypergraph in question has 64 terms rather than 16, so a direct repetition of Appendix A is not realistic; a new structural input is needed.

Question 7.1. What is R_4 ? Is it 22?

7.2. The extremality conjecture. Since $R_2 = 4 = 3 \cdot 2^1 - 2$ and $R_3 = 10 = 3 \cdot 2^2 - 2$, both equal to the size of the Chiarelli–Hatami–Saks construction, it is natural to conjecture that the construction is always extremal.

Conjecture 7.2. $R_d = 3 \cdot 2^{d-1} - 2$ for every $d \geq 2$; equivalently, $C_{44} = 3/2$.

Conjecture 7.2 would close entry 44a of [4] completely. We stress that [1] supply the construction but do not state this conjecture; the evidence for it is the two exact values above, and it is far from the known upper bound $4.394 \cdot 2^d$.

One natural line of attack is the recursion behind the construction: Ξ_d is built from two copies of Ξ_{d-1} and two new selector variables, giving $R_d \geq 2R_{d-1} + 2$. A matching upper bound

$$R_d \leq 2R_{d-1} + 2 \tag{7}$$

would prove Conjecture 7.2 by induction from $R_2 = 4$. The link structure of Section 3 is the natural tool: Remark 3.2 says that the derivative at any variable of minimal influence 2^{1-d} is a character times the indicator of an affine subspace of codimension $d - 1$, *without* any homogeneity hypothesis. What is missing is the corresponding structure for variables of the next influence levels $2^{2-d}, 2^{3-d}, \dots$ — in Ξ_d these are exactly the selector variables, and they are the variables that (7) would have to control.

Question 7.3. Describe the derivatives $D_i f$ of a degree- d Boolean function at variables of influence 2^{2-d} . Does a structure theorem at the first few influence levels imply (7)?

7.3. Uniqueness of the extremal degree-3 function. The exhaustive search of Section 6 found exactly one orbit at $n = 10$, namely Ξ_3 , under the group generated by permutations of variables, negation of variables and the global sign. So, computationally, the extremal degree-3 function is unique up to those symmetries. We have no proof of this by hand.

Question 7.4. Prove without computer search that Ξ_3 is, up to permutation and negation of variables and global sign, the unique degree-3 Boolean function with 10 relevant variables. More generally, is the extremal function unique at every d ?

7.4. On the method. The reusable content of this paper is Lemma 3.1 together with Remark 3.2: a $\{-1, 0, 1\}$ -valued function whose Fourier mass is as small as granularity permits is forced to be a character times an affine-subspace indicator. This is a statement about sparse granular functions, with no degree hypothesis, and it is what converts an influence-counting argument into a rigid combinatorial one. At $d = 3$ the resulting objects are closed triangulated surfaces, which is why a complete classification is possible; for general d the same device yields cross-polytopes and a clean disjointness contradiction, which is exactly enough to strip one variable off the Nisan–Szegedy bound, but not obviously more.

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APPENDIX A. THE ELEVEN-VARIABLE CASE ANALYSIS

This appendix proves Theorem 5.5. Notation is that of Section 5: $f: \{-1, 1\}^{11} \rightarrow \{-1, 1\}$ has degree at most 3 with all 11 variables relevant, $n_S = 4\hat{f}(S) \in \mathbb{Z}$, $\sum_S n_S^2 = 16$, $m_i = \sum_{S \ni i} n_S^2$, and by (6) we have $e + \delta = 4$ with $\delta \in \{4, 2, 0\}$. We write “ $m = 4$ vertex” for a vertex of mass 4, and use the link shapes (L1)–(L4) of Lemma 5.3(a) throughout. We refer to a term S with $|S| = 3$ as a *cubic*, $|S| = 2$ as a *quadratic*, $|S| = 1$ as a *linear* term.

A.1. The case $\delta = 4$: all masses are 4. Here every $m_i = 4$, so every link is (L1), (L2), (L3) or (L4). The lower-order weight 4 is partitioned as one of $\{1, 1, 1, 1\}$, $\{4\}$, $\{2, 1, 1\}$, $\{2, 2\}$, $\{3, 1\}$, with contributions as listed in Section 5.3.

Partition $\{4\}$. A quadratic term with $|n_S| = 2$. By Lemma 5.3(d) its endpoints would need mass 8, contradicting $m_i = 4$. ×

Partition $\{2, 2\}$. Two linear terms $\{i\}$ and $\{j\}$, and no quadratics. The link of i contains $\emptyset$, and there are no quadratics, so it is (L4): the cubics at i are $\{i, a, b\}$, $\{i, b, c\}$, $\{i, a, c\}$. Likewise j has an (L4) link.

If $j \in \{a, b, c\}$, say $j = a$, then the link of a consists of $\emptyset$ together with a triangle containing the pairs $\{i, b\}$ and $\{i, c\}$, which forces the cubic $\{a, b, c\}$. But then the link of b contains $\{i, a\}$, $\{i, c\}$, $\{a, c\}$, a triangle, although b has no lower-order term and $m_b = 4$, so the link of b must be a C_4 — and a C_4 contains no triangle. ×

Hence $j \notin \{a, b, c\}$, and a, b, c have (L1) links. The link of a contains $\{i, b\}$ and $\{i, c\}$, which completes to the 4-cycle i – b – w – c , giving the cubics $\{a, b, w\}$ and $\{a, c, w\}$; the same argument at b gives $\{a, b, w'\}$, $\{b, c, w'\}$ and at c gives $\{a, c, w''\}$, $\{b, c, w''\}$, and the shared cubics force

$w = w' = w''$. Then the link of w contains the triangle $\{a, b\}, \{a, c\}, \{b, c\}$, so w is (L4), so $w = j$ (the only other vertex with a linear term). The terms

$$\{i\}, \{j\}, \{i, a, b\}, \{i, b, c\}, \{i, a, c\}, \{j, a, b\}, \{j, b, c\}, \{j, a, c\}$$

exhaust the vertices i, j, a, b, c (each of mass 4) and carry weight 8; the remaining weight 8 lives on the other 6 variables. So f is a sum of two non-constant functions on disjoint variable sets, impossible by Lemma 3.5. $\times$

Partition $\{3, 1\}$. A constant term and one quadratic $\{i, a\}$. The link of i contains the singleton $\{a\}$, so it is (L2) or (L3) by Lemma 5.3(a); but (L2) needs a linear term at i , and there is none (the weight is spent on the constant and the quadratic), while (L3) needs two quadratics at i , and there is only one quadratic in total. $\times$

Partition $\{2, 1, 1\}$. One linear term $\{i\}$ and two quadratics. The link of i contains $\emptyset$, so it is (L2) or (L4).

If it is (L2) = $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$: the two quadratics are $\{i, a\}$ and $\{i, b\}$, and $\{i, a, b\}$ is a cubic. The link of a contains the singleton $\{i\}$ (from the quadratic $\{i, a\}$), so a is (L2) or (L3). It cannot be (L2), which would need a linear term $\{a\}$, while the only linear term is $\{i\}$. So it is (L3), of the form $\{\{p\}, \{q\}, \{p, r\}, \{q, r\}\}$ with $\{i\} = \{p\}$; the pair $\{i, b\}$ (coming from the cubic $\{i, a, b\}$) must be $\{p, r\}$, so $r = b$, and then the fourth member $\{q, r\}$ requires a quadratic $\{a, q\}$ with $q \neq i$. But the only quadratics are $\{i, a\}$ and $\{i, b\}$. $\times$

If it is (L4): no quadratic touches i . So every endpoint u of a quadratic has $u \neq i$, has a singleton in its link, and has no linear term of its own; hence u is (L3) and therefore lies on exactly two quadratics. The two quadratics would then form a 2-regular graph with 2 edges, which does not exist. $\times$

Partition $\{1, 1, 1, 1\}$. Four unit quadratics forming a graph Q , with no linear and no constant term. Every endpoint u of a quadratic has a singleton in its link, hence is (L3) (it cannot be (L2), which needs a linear term), hence lies on exactly two quadratics. So Q is a 2-regular graph with 4 edges, i.e. a 4-cycle $i-a-j-b-i$. By (L3) at each of i, a, j, b , the two cubics at i are $\{i, a, c_i\}, \{i, b, c_i\}$ for some c_i , and similarly at a, j, b .

Let $S_0 = \{i, a, j, b\}$ and let N be the set of cubics meeting S_0 . Each member of N meets S_0 in 2 or 3 vertices, since it contains one of i, a, j, b together with a Q -neighbour of that vertex. Each of i, a, j, b lies in exactly two cubics, so $\sum_{T \in N} |T \cap S_0| = 8$. The seven vertices outside S_0 all have mass 4, hence

$$28 = \sum_{T \in N} |T \setminus S_0| + 3M,$$

where M is the number of cubics disjoint from S_0 , while counting all 16 terms gives $16 = 4 + |N| + M$. Writing $k := \#\{T \in N : |T \setminus S_0| = 1\}$ (the others have $|T \setminus S_0| = 0$), we get $3(12 - |N|) = 28 - k$, so

$$k = 3|N| - 8 \leq |N|,$$

which forces $|N| \in \{3, 4\}$.

Sub-case $|N| = 4$. All four cubics of N have exactly two vertices in S_0 and one outside. Take $\{i, a, c\} \in N$. The vertex c has an (L1) link containing the pair $\{i, a\}$, so the 4-cycle in $\text{lk}(c)$ is $i-a-y-z-i$, giving cubics $\{c, a, y\}, \{c, y, z\}, \{c, z, i\}$. Now a 's only other cubic is $\{a, j, c_a\}$, so $y = j$ and $c_a = c$; and i 's only other cubic is $\{i, b, c_i\}$, so $z = b$ and $c_i = c$; and $\{c, j, b\}$ is a cubic. Thus

$$N = \{\{c, a, i\}, \{c, a, j\}, \{c, j, b\}, \{c, b, i\}\},$$

$m_c = 4$ is exhausted, and i, a, j, b are exhausted too (two quadratics and two cubics each). Hence no other term meets $\{i, a, j, b, c\}$, and $f = g(x_i, x_a, x_j, x_b, x_c) + h(\text{the other 6 variables})$ with both parts non-constant, impossible by Lemma 3.5. $\times$

Sub-case $|N| = 3$. Then $k = 1$: two cubics lie inside S_0 and one meets it in an edge. Two 3-subsets of the 4-cycle's vertex set share two vertices, so up to the symmetry of the 4-cycle they are $\{i, a, j\}, \{i, a, b\}$ (sharing the Q -edge $\{i, a\}$) or $\{i, a, j\}, \{j, b, i\}$ (sharing the diagonal $\{i, j\}$).

- First: $\text{lk}(i) = \{\{a\}, \{b\}, \{a, j\}, \{a, b\}\}$, which is not of the form $\{\{p\}, \{q\}, \{p, r\}, \{q, r\}\}$ required by (L3). $\times$
- Second: $\text{lk}(a)$ contains the pair $\{i, j\}$ (from the cubic $\{i, j, a\}$); but an (L3) link at a has the form $\{\{p\}, \{q\}, \{p, r\}, \{q, r\}\}$ with $\{p, q\} = \{i, j\}$ and $r \notin \{i, j\}$, so $\{i, j\}$ would have to be $\{i, r\}$, forcing $r = j$. $\times$

So $\delta = 4$ is impossible.

A.2. The case $\delta = 2$: one vertex of mass 6. Here exactly one vertex v has $m_v = 6$ and the other ten have mass 4. The lower-order weight 2 is either one linear term ± 1 or two unit quadratics.

One linear term $\{l\}$. The link of l contains $\emptyset$. There are no quadratics, so every vertex other than l has a link made of pairs only: a C_4 if its mass is 4, and (for mass 6) a simple graph with 6 edges and all degrees even, i.e. C_6 , two disjoint triangles, or a bowtie (two triangles sharing a vertex).

Suppose $m_l = 4$. Then $\text{lk}(l)$ is (L4), giving cubics $\{l, a, b\}, \{l, b, c\}, \{l, a, c\}$.

Assume first $v \notin \{a, b, c\}$, so a, b, c have C_4 links. The link of a contains $\{l, b\}$ and $\{l, c\}$, which completes to $l-b-w-c$, giving cubics $\{a, b, w\}, \{a, c, w\}$; similarly at b and at c , and the shared cubics force a common w , so the cubics $\{a, b, w\}, \{a, c, w\}, \{b, c, w\}$ all exist. Then $\text{lk}(w)$ contains the triangle abc , so it is not a C_4 ; hence $w = v$ (and $w \neq l$, since l 's link is already exhausted). So $\text{lk}(v)$ contains the triangle abc and $m_v = 6$: thus $\text{lk}(v)$ is two triangles $abc \cup xyz$, or a bowtie. A bowtie containing abc has its centre in $\{a, b, c\}$, say a , giving $\lambda_{va} = 4$ and hence a vertex of degree 4 in $\text{lk}(a)$, which is not a C_4 — contradiction. So $\text{lk}(v) = abc \cup xyz$, with cubics $\{v, x, y\}, \{v, y, z\}, \{v, x, z\}$. Now x, y, z have C_4 links, and the same completion argument yields cubics $\{x, y, t\}, \{x, z, t\}, \{y, z, t\}$ for a common t , so $\text{lk}(t)$ contains a triangle; but $t \notin \{v, l\}$ (both of those links are already exhausted), so t has a C_4 link containing a triangle. $\times$

Assume now $v \in \{a, b, c\}$, say $v = a$. Then b and c have C_4 links. The link of b contains $\{l, a\}$ and $\{l, c\}$, giving cubics $\{a, b, w\}, \{b, c, w\}$; the link of c contains $\{l, a\}$ and $\{l, b\}$, giving $\{a, c, w'\}, \{b, c, w'\}$; the shared cubic $\{b, c, \cdot\}$ forces $w = w'$. So $\text{lk}(w)$ contains the triangle abc . Here $w \notin \{l, a\}$ (taking $w = l$ would repeat the term $\{l, a, b\}$, and $w = a$ is not allowed since $\{a, a, b\}$ is not a set), so w has a C_4 link containing a triangle. $\times$

Hence $m_l = 6$, i.e. $l = v$. Then $\text{lk}(v)$ is $\emptyset$ together with five pairs having all degrees even, i.e. a 5-cycle $a_1 a_2 a_3 a_4 a_5$, with cubics $\{v, a_k, a_{k+1}\}$ (indices mod 5). Each a_k has an (L1) link containing $\{v, a_{k-1}\}$ and $\{v, a_{k+1}\}$, which completes to a 4-cycle $v-a_{k+1}-w_k-a_{k-1}-v$, giving cubics $\{a_k, a_{k+1}, w_k\}$. If w_k were another a_p , some a 's link would contain a triangle (for instance $w_1 = a_3$ makes $\text{lk}(a_2)$ contain $\{v, a_1\}, \{v, a_3\}, \{a_1, a_3\}$), impossible inside a C_4 . So w_k lies outside $\{v, a_1, \dots, a_5\}$. The cubic $\{a_k, a_{k+1}, w_k\}$ is one of a_{k+1} 's two non- v cubics, so $w_{k+1} = w_k$; hence all w_k are equal to a single vertex w , which then lies in five cubics, i.e. $m_w = 5$, contradicting Lemma 5.1. $\times$

Two quadratics. An endpoint u of a quadratic with $m_u = 4$ has a singleton in its link, so its link is (L2) or (L3); (L2) needs a linear term, and there is none, so it is (L3), which needs two quadratics at u : say $\{u, p\}$ and $\{u, q\}$ with $p \neq q$. If $p \neq v$ then p is a mass-4 endpoint too and would need two quadratics at p , but both quadratics already meet u and there are only two in total — so $p = v$, and likewise $q = v$, contradicting $p \neq q$. So no mass-4 vertex is an endpoint of a quadratic, which means each quadratic has both endpoints equal to v , absurd. $\times$

So $\delta = 2$ is impossible.

A.3. The case $\delta = 0$: a homogeneous cubic. Now $f = \frac{1}{4} \sum_S n_S \chi_S$ is supported on triples, with $\sum_S n_S^2 = 16$, and the degree sequence (m_i) is either $(8, 4^{10})$ or $(6, 6, 4^9)$. No $|n_S| = 2$ occurs: by Lemma 5.3(d) all three vertices of such a term would need mass 8, and there is at most one vertex of mass 8. So $\mathcal{F}$ is a 3-uniform hypergraph with 16 triples. Links are: a C_4 when $m = 4$ (Lemma 5.3(a)); for $m = 6$, a simple graph with 6 edges and all degrees even, i.e. C_6 , two disjoint triangles, or a bowtie; for $m = 8$, a simple graph with 8 edges and all degrees even.

A.3.1. Sub-case $(8, 4^{10})$. Let v be the vertex with $m_v = 8$. If some x had $\lambda_{vx} = 4$, then v would have degree 4 in $\text{lk}(x)$, so $\text{lk}(x)$ would not be a C_4 , contradicting $m_x = 4$ (and $x \neq v$, since x is a vertex of $\text{lk}(v)$). Hence $\text{lk}(v)$ is 2-regular: C_8 , $C_3 \cup C_5$, or $C_4 \cup C_4$; and every pair lies in 0 or 2 triples, so $|\mathcal{F}|$ is a closed pseudo-manifold, with all links single cycles except possibly at v .

$\text{lk}(v) = C_8$. Then $|\mathcal{F}|$ is a closed surface with $V = 11$, $E = 24$, $F = 16$, so $\chi = 3$. Consider its components. A closed surface with all degrees 4 except at most one vertex of degree 8 has degree sum $4V' + 4 \cdot \mathbf{1}\{v \in \text{it}\}$. A sphere needs degree sum $6V' - 12$, giving either $V' = 6$ (the octahedron, not containing v) or $6V' - 12 = 4V' + 4$, i.e. $V' = 8$ — a sphere on 8 vertices with degrees $(8, 4^7)$, which is impossible since v would need 8 neighbours among 7 other vertices. A projective plane needs $6V' - 6$, giving $V' = 3$ or $V' = 5$, both below the minimum of 6 vertices. And $\chi' \leq 0$ is excluded by Fact 5.4. So $\chi = 3$ cannot be realised as a sum of component characteristics. $\times$

$\text{lk}(v) = C_4 \cup C_4$. Split v into two vertices v_1, v_2 , each of degree 4. The result is a closed surface with 12 vertices all of degree 4, so $\chi = 12 - 24 + 16 = 4$ and, exactly as in Step 3 of Theorem 4.1, it is two octahedra. If v_1 and v_2 lay in the same octahedron — this sub-case is in fact empty, since the two C_4 's of $\text{lk}(v)$ are vertex-disjoint while two vertices of an octahedron share two or four neighbours — the other octahedron would use 6 variables disjoint from everything else and f would be a disjoint sum, impossible by Lemma 3.5. Otherwise the two octahedra are glued at the identified vertex v . Write

$$f = g_1(x_v, y) + g_2(x_v, z), \quad y, z \text{ disjoint 5-tuples.}$$

Fixing $x_v = 1$ gives $g_1(1, y) + g_2(1, z) \in \{\pm 1\}$ for all y, z , so by Lemma 3.5 one of them is constant. But $g_1(1, y)$ is $\frac{1}{4}$ times the sum of the eight face-characters of the first octahedron with $x_v = 1$ substituted, which are eight distinct non-constant monomials in y (four quadratic and four cubic), so $g_1(1, \cdot)$ is not constant; and the same holds for $g_2(1, \cdot)$. $\times$

$\text{lk}(v) = C_3 \cup C_5$. Split v into v_1 of degree 3 and v_2 of degree 5: a closed surface with 12 vertices, degrees $(3, 5, 4^{10})$, and $\chi = 4$. Components with $\chi' \leq 1$ are excluded: $\chi' \leq 0$ by Fact 5.4; and a projective plane needs degree sum $6V' - 6$, which with all degrees at most 5 forces $V' \leq 6$ (and a triangulated projective plane has at least 6 vertices, by Step 3 of Theorem 4.1), while the only 6-vertex projective plane has all six degrees equal to 5, whereas we have a single degree-5 vertex. Hence both components are spheres, with degree sum $6V' - 12$ and degrees drawn from $\{3, 4, 5\}$:

- containing both the 3 and the 5: $8 + 4a = 6(a + 2) - 12$ gives $a = 4$, a 6-vertex sphere with degrees $(5, 4, 4, 4, 4, 3)$. The degree-5 vertex is adjacent to all the others, its link is a 5-cycle, and the remaining 3 faces triangulate that pentagon with two diagonals; any two diagonals of a pentagon share a vertex or cross, so some vertex receives diagonal degree 2 or the triangulation is invalid, and the required degree pattern $(4, 4, 4, 4, 3)$ on the rim, i.e. diagonal degrees $(1, 1, 1, 1, 0)$, cannot occur.
- containing the 3 but not the 5: $3 + 4a = 6(a + 1) - 12$ gives $a = 4.5$, not an integer.
- containing the 5 but not the 3: $5 + 4a = 6(a + 1) - 12$ gives $a = 5.5$, not an integer.

×

A.3.2. *Sub-case $(6, 6, 4^9)$.* Let v and v' be the two vertices of mass 6.

Some link is a bowtie. Say $\text{lk}(v)$ is a bowtie with centre w . Then $\lambda_{vw} = 4$, so v has degree 4 in $\text{lk}(w)$, which forces $m_w = 6$ and $\text{lk}(w)$ a bowtie too, so $w = v'$ and, symmetrically, $\text{lk}(v')$ is a bowtie centred at v . The four common triples are $\{v, v', a_k\}$, $k = 1, \dots, 4$. The bowtie at v adds two further triples $\{v, a_1, a_2\}, \{v, a_3, a_4\}$ (for some pairing of the a_k), and the one at v' adds $\{v', a_i, a_j\}, \{v', a_k, a_l\}$ for some pairing. If the two pairings coincide, then $\text{lk}(a_1)$ contains $\{v, v'\}, \{v, a_2\}, \{v', a_2\}$, a triangle, hence is not a C_4 . Otherwise, say v pairs $(a_1 a_2)(a_3 a_4)$ while v' pairs $(a_1 a_3)(a_2 a_4)$: then $\text{lk}(a_1)$ contains $\{v, v'\}, \{v, a_2\}, \{v', a_3\}$, and being a C_4 it forces the fourth triple $\{a_1, a_2, a_3\}$; but then $\text{lk}(a_2) = \{\{v, v'\}, \{v, a_1\}, \{v', a_4\}, \{a_1, a_3\}\}$ has a_3 and a_4 of degree 1, so it is not a C_4 . ×

No link is a bowtie. Then every pair lies in 0 or 2 triples and $|\mathcal{F}|$ is a closed pseudo-manifold, singular only at v or v' and only when the link there is two triangles.

If both links are cycles, $|\mathcal{F}|$ is a closed surface with $\chi = 11 - 24 + 16 = 3$. Component degree sums are $4V' + 2 \cdot \mathbf{1}\{v\} + 2 \cdot \mathbf{1}\{v'\}$. Spheres: $V' = 6$ (containing neither v nor v' , the octahedron); or, containing exactly one of them, $4V' + 2 = 6V' - 12$, i.e. $V' = 7$ with degrees $(6, 4^6)$ — here the degree-6 vertex is adjacent to all 6 others, its link is a hexagon, and the remaining 4 faces triangulate the hexagon with 3 diagonals, so the six rim vertices have degrees $3 + d_i$ with $\sum_i d_i = 6$; all degrees equal to 4 needs $d_i = 1$ for every i , a perfect matching of the hexagon by pairwise non-crossing diagonals, which does not exist (every hexagon triangulation has an ear). Containing both v and v' : $4V' + 4 = 6V' - 12$, i.e. $V' = 8$ with degree sum 36, two degree-6 vertices and six degree-4 vertices — this is not immediately absurd, but then the remaining components would have to be octahedra on the remaining 3 vertices, which is impossible; and $V' = 11$ as a single component gives $\chi = 3 \neq 2$. Projective planes: $6V' - 6 = 4V' + 2s$ with $s \in \{0, 1, 2\}$ gives $V' = 3, 4, 5$, all below 6. And $\chi' \leq 0$ is excluded by Fact 5.4. So $\chi = 3$ is not realisable. ×

If $\text{lk}(v)$ is two triangles, split v into v_1, v_2 of degree 3 each. If $\text{lk}(v')$ is a cycle we obtain 12 vertices with degrees $(3, 3, 6, 4^9)$ and $\chi = 4$; both components are then spheres, since $\chi' \leq 0$ is excluded by Fact 5.4 and a projective plane would need $6V' - 6 \leq 6 + 4(V' - 1)$, i.e. $V' \leq 4 < 6$. The sphere containing v' satisfies $6 + 3a + 4b = 6(1 + a + b) - 12$, i.e. $3a + 2b = 12$ with $a \leq 2$: the solution $(a, b) = (2, 3)$ gives a 6-vertex sphere with v' of degree 6, impossible as there are only 5 other vertices; and $(a, b) = (0, 6)$ gives the $(6, 4^6)$ sphere excluded above. ×

If $\text{lk}(v')$ is also two triangles, both split, giving degrees $(3, 3, 3, 3, 4^9)$ with $\chi = 5$: odd, hence impossible as a sum of sphere characteristics; projective planes with all degrees at most 4 need $6V' - 6 \leq 4V'$, i.e. $V' \leq 3 < 6$; and $\chi' \leq 0$ is excluded by Fact 5.4. ×

So $\delta = 0$ is impossible, and with Sections A.1 and A.2 all cases are exhausted. This proves Theorem 5.5. □

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